

# Super-sample covariance of windowed power-spectrum multipoles in `thecov`

Redshift-space responses, the local average of a realised normalisation,  
and long-mode variances from tripolar window pair counts

Formalism note, branch `thecov2` — October 2026. Code: `thecov/ssc.py`; tests: `tests/test_ssc.py`; DESI check: `desi_validation/ssc_check.py`

**Summary.** Density fluctuations on scales comparable to or larger than the survey modulate the measured power spectrum coherently at all  $k$ . In the DESI DR2 LRG1 mocks this term is measured directly in the off-diagonal elements of the mock covariance, where the windowed Gaussian covariance vanishes: it is  $A_{\ell\ell'} P_0(k) P_0(k')$  with coherent amplitudes  $\sigma_{P_0} = 0.51\%, 0.66\%$  and  $\sigma_{P_2}/P_0 = 0.94\%, 1.22\%$  (NGC, SGC), nearly uncorrelated, and it accounts for the quadrupole-amplitude error of the Gaussian covariance. This note derives the super-sample covariance (SSC) that `thecov` now computes: tree-level redshift-space responses of each multipole to a long mode, split by the long mode's orientation to the local line of sight; the local-average effect of an estimator whose  $\alpha$  and normalisation are computed from each realisation; and the long-mode variances as one-dimensional integrals of the linear correlation function against the tripolar window functions `thecov` already measures from pair counts. Nothing is fitted to the mocks.

## 1 Estimator and long mode

**Estimator.** Galaxies and randoms with total weights  $w$ ,  $\alpha = \sum_g w_g / \sum_r w_r$ , the weighted field  $F(\mathbf{x}) = w(n_g - \alpha n_r)$  and, as in `pypower/jaxpower`,

$$\hat{P}_\ell(k) = \frac{2\ell + 1}{\text{norm}} \int \frac{d\Omega_k}{4\pi} |\tilde{F}(\mathbf{k})|^2 \mathcal{L}_\ell(\hat{\mathbf{k}} \cdot \hat{\mathbf{n}}) - \text{shot noise}, \quad \text{norm} = \alpha \sum_c D_c R_c / V_c, \quad (1)$$

with the local line of sight  $\hat{\mathbf{n}} = \hat{\mathbf{x}}$  and the normalisation computed *for each realisation* from its own data  $D_c$  and randoms  $R_c$  in cells of  $10h^{-1}\text{Mpc}$ . Write  $m(\mathbf{x}) = \bar{n}(\mathbf{x}) \langle w \rangle(\mathbf{x})$  for the mean weighted density; the clustering window of the Gaussian covariance is  $W = m^2$  (or its pair-averaged version, see `window_kernel.typ`), with  $\int W = I$ , and the mean of the estimator is  $P(k)I_k / \text{norm}$ .

**Long mode.** Split the matter density into a long mode  $\delta_L$ , with wavenumbers  $q$  below the resolved  $k$  and up to the survey scale, and short modes. To first order in  $\delta_L$  the short-mode power at position  $\mathbf{x}$  is shifted by

$$\Delta P(\mathbf{k}\mathbf{x}) = \int \frac{d^3q}{(2\pi)^3} \delta_L(\mathbf{q}) e^{i\mathbf{q}\cdot\mathbf{x}} R(\mathbf{k}, \hat{\mathbf{q}}; \hat{\mathbf{n}}), \quad (2)$$

where  $R$  is the response of the redshift-space galaxy power spectrum to a long matter mode of unit amplitude along  $\hat{\mathbf{q}}$ .

## 2 Responses

### 2.1 Squeezed limit of the tree-level trispectrum

With the redshift-space perturbation-theory kernels  $Z_1, Z_2$  (Scoccimarro, Couchman & Frieman 1999; local line of sight  $\hat{\mathbf{n}}$ ),

$$Z_1(\mathbf{k}) = b_1 + f\mu^2, \quad (3)$$

$$Z_2(\mathbf{k}_1, \mathbf{k}_2) = b_1 F_2 + f \mu^2 G_2 + \frac{f \mu k}{2} \left[ \frac{\mu_1}{k_1} Z_1(\mathbf{k}_2) + \frac{\mu_2}{k_2} Z_1(\mathbf{k}_1) \right] + \frac{b_2}{2} + \frac{b_{s^2}}{2} S_2, \quad (4)$$

with  $\mathbf{k} = \mathbf{k}_1 + \mathbf{k}_2$ ,  $\mu = \hat{\mathbf{k}} \cdot \hat{\mathbf{n}}$  and  $S_2 = (\hat{\mathbf{k}}_1 \cdot \hat{\mathbf{k}}_2)^2 - 1/3$ . The second-order field contains  $2Z_2(\mathbf{k} - \mathbf{q}, \mathbf{q})\delta(\mathbf{k} - \mathbf{q})\delta_L(\mathbf{q})$ , and its correlation with the linear field gives the response

$$R(\mathbf{k}, \hat{\mathbf{q}}) = \lim_{q \rightarrow 0} 2[Z_2(\mathbf{k}, -\mathbf{q})Z_1(\mathbf{k})P(k) + Z_2(-\mathbf{k} + \mathbf{q}, -\mathbf{q})Z_1(\mathbf{k} - \mathbf{q})P(|\mathbf{k} - \mathbf{q}|)]. \quad (5)$$

The two terms are each  $\mathcal{O}(1/q)$ : the bulk flow displaces short modes without changing their power. These pieces are odd in  $\mathbf{q}$  and cancel; the finite remainder contains growth ( $F_2$ ,  $G_2$ ), dilation ( $P(|\mathbf{k} - \mathbf{q}|)$  to first order in  $\mathbf{q}$ , i.e.  $dP/d \ln k$ ), the bias responses  $b_2$ ,  $b_{s^2}$ , and the redshift-space terms (the long mode's Kaiser and velocity-gradient effect on the short modes).  $P$  is the linear matter power spectrum and  $\delta_L$  the linear matter long mode.

## 2.2 Multipoles and the long-mode orientation

The estimator averages over  $\hat{\mathbf{k}}$  with weight  $\mathcal{L}_\ell(\hat{\mathbf{k}} \cdot \hat{\mathbf{n}})$ . After that average,  $R$  depends on  $\hat{\mathbf{q}}$  only through  $\nu = \hat{\mathbf{q}} \cdot \hat{\mathbf{n}}$ , so

$$R_\ell(k; \nu) = \frac{2\ell + 1}{2} \int_{-1}^1 d\mu \mathcal{L}_\ell(\mu) \langle R \rangle_\varphi = \sum_n R_\ell^{(n)}(k) \mathcal{L}_n(\nu), \quad (6)$$

$$R_\ell^{(n)}(k) = a_{\ell n} P(k) + c_{\ell n} \frac{dP}{d \ln k}, \quad (7)$$

with numbers  $a_{\ell n}$ ,  $c_{\ell n}$  that depend on  $(b_1, b_2, b_{s^2}, f)$ . `response_coefficients` evaluates Equation 5 on a quadrature grid in  $(\mu, \nu, \varphi)$  and takes the squeezed limit as the average of  $q = \pm \varepsilon$ , which removes the odd  $1/\varepsilon$  terms exactly and leaves an  $\mathcal{O}(\varepsilon^2)$  error ( $10^{-8}$  for  $\varepsilon = 10^{-4}$ ). Gauss–Legendre in  $\mu$  and  $\nu$  and a uniform grid in  $\varphi$  are exact for the polynomials that occur.

**Checks** (tested). Real space,  $b_1 = 1$ :  $R_0^{(0)} = (47/21)P - (1/3)dP/d \ln k$  and  $R_2^{(2)} = (2/3)(8/7)P - (2/3)dP/d \ln k$ , the isotropic and tidal (separate-universe) responses, and nothing else. Galaxy bias adds  $2b_1 b_2 P$  to  $R_0^{(0)}$  and  $(4/3)b_1 b_{s^2} P$  to  $R_2^{(2)}$ . In redshift space  $R_\ell^{(4)} = 0$ : only  $n = 0, 2$  occur. For  $b_1 = 2.2$ ,  $f = 0.76$ ,  $b_2 = 0.28$  (Lazeyras et al. 2016),  $b_{s^2} = -4/7(b_1 - 1)$ :

| $\ell$ | $a_{\ell 0}/c_{\ell 0}$ | $a_{\ell 2}/c_{\ell 2}$ |
|--------|-------------------------|-------------------------|
| 0      | 19.02/−2.62             | 3.76/−1.54              |
| 2      | 12.43/−2.22             | 6.02/−7.30              |
| 4      | 1.31/−0.28              | 2.46/−1.38              |

For comparison, the Kaiser monopole and quadrupole are  $6.07P$  and  $2.56P$ . Two consequences:  $P_2$  responds to the isotropic mode ( $a_{20}$ , mostly through the Kaiser term) as well as to the line-of-sight component  $n = 2$ , and its largest dilation coefficient sits in  $c_{22}$ . These are the two “amplitude” directions found in the mocks.

## 3 Response of the estimator: beat coupling and local average

With the long mode,  $n_g = \bar{n}(1 + \delta_g)$  where  $\delta_g$  is referenced to the global mean. The short-mode part of  $|\tilde{F}|^2$  is the local power of  $\delta_g$  weighted by the clustering window, so it gains  $\int W(\mathbf{x}) \Delta P(\mathbf{k}\mathbf{x})$ : the *beat coupling* (BC). The constant  $\alpha$  only shifts the  $k \approx 0$  modes. The normalisation, however, follows the realisation:

$$\text{norm} \propto \alpha \sum_c D_c R_c \propto (1 + \bar{\delta}^M)(1 + \bar{\delta}^W), \quad \bar{\delta}^w = \frac{\int w(\mathbf{x}) \delta_{g,L}(\mathbf{x})}{\int w}, \quad (8)$$

with  $M = m$  (the average in  $\alpha = \sum_g w / \sum_r w$ ) and  $W = m^2$  (the average of the data in  $D_c R_c$ ).  $\delta_{g,L} = (b_1 + f\nu^2)\delta_L$  is the observed long mode. Hence, to first order,

$$\Delta \hat{P}_\ell(k) = d_k \sum_n R_\ell^{(n)}(k) D_n^W - \hat{P}_\ell(k) \sum_n g_n (D_n^W + D_n^M), \quad (9)$$

$$D_n^w = \frac{1}{\int w} \int d^3 \mathbf{x} w(\mathbf{x}) \int \frac{d^3 q}{(2\pi)^3} \delta_L(q) e^{i\mathbf{q} \cdot \mathbf{x}} \mathcal{L}_n(\hat{\mathbf{q}} \cdot \hat{\mathbf{x}}), \quad g_0 = b_1 + \frac{f}{3}, \quad g_2 = \frac{2f}{3}. \quad (10)$$

The second term is the *local average* (LA). For a uniform window and an isotropic long mode in real space it is  $-2b_1 P$ , and Equation 9 reduces to the familiar  $(R-2)$ . Here it splits into an  $m$ -weighted part from  $\alpha$  and an  $m^2$ -weighted part from the data in `norm`. Those two parts differ when  $m$  varies across the survey. The DESI LRG1 mocks show this cancellation: the measured slope  $\partial \hat{P}_0 / \partial \delta_{\text{norm}} \approx (0.1 \text{ to } 0.25) \bar{P}_0$  is small and positive, as  $(R-2)/2$  predicts, not  $-\bar{P}_0$ .

**Dilution.**  $d_k = I_k / \text{norm}$  converts the response of the true power into that of the measured, window-diluted spectrum. For footprints with fine veto masks it must be the pair-averaged window's  $I_k$ , which is  $\approx 1$  for DESI LRG1. The local  $\int m^2 / \text{norm} = 1.22$  would inflate the BC variance by 1.5. In the LA term  $\hat{P}_\ell$  is the measured spectrum itself. That is the model when the covariance's model is masked (the mock mean), and the model times  $d_k$  otherwise. Setting `local_average=False` describes a normalisation fixed in advance.

## 4 Long-mode variances from the window pair counts

The covariance of the projections in Equation 9 is

$$\sigma_{(wn)(w'n')}^2 = \langle D_n^w D_{n'}^{w'} \rangle = \frac{1}{I_w I_{w'}} \int d^3 \mathbf{x} d^3 \mathbf{x}' w(\mathbf{x}) w'(\mathbf{x}') \int \frac{d^3 q}{(2\pi)^3} P(q) e^{-i\mathbf{q} \cdot \mathbf{s}} \mathcal{L}_n(\hat{\mathbf{q}} \cdot \hat{\mathbf{x}}) \mathcal{L}_{n'}(\hat{\mathbf{q}} \cdot \hat{\mathbf{x}'}), \quad (11)$$

with  $\mathbf{s} = \mathbf{x}' - \mathbf{x}$ . Expanding the plane wave and both Legendre polynomials in spherical harmonics, the  $\hat{\mathbf{q}}$  integral is a Gaunt coefficient. The remaining angular dependence on  $(\hat{\mathbf{x}}, \hat{\mathbf{x}'}, \hat{\mathbf{s}})$  is the real tripolar harmonic

$$S_{nn'\lambda}(\hat{\mathbf{x}}, \hat{\mathbf{x}'}, \hat{\mathbf{s}}) = \sum_{mm'M} \begin{pmatrix} n & n' & \lambda \\ m & m' & M \end{pmatrix} Y_{nm}^*(\hat{\mathbf{x}}) Y_{n'm'}^*(\hat{\mathbf{x}'}) Y_{\lambda M}^*(\hat{\mathbf{s}}) \quad (12)$$

used by `thecov`'s window pair counts,

$$Q_{nn'\lambda}^{ww'}(s) = \int d\Omega_s \int d^3 \mathbf{x} w(\mathbf{x}) w'(\mathbf{x} + \mathbf{s}) S_{nn'\lambda}(\hat{\mathbf{x}}, \hat{\mathbf{x}'}, \hat{\mathbf{s}}). \quad (13)$$

The result is a sum of one-dimensional integrals,

$$\sigma_{(wn)(w'n')}^2 = \frac{(4\pi)^{3/2}}{I_w I_{w'}} \sum_\lambda (-1)^{\lambda/2} \sqrt{\frac{2\lambda+1}{(2n+1)(2n'+1)}} \begin{pmatrix} n & n' & \lambda \\ 0 & 0 & 0 \end{pmatrix} \int s^2 ds \xi_\lambda(s) Q_{nn'\lambda}^{ww'}(s), \quad (14)$$

$$\xi_\lambda(s) = \int \frac{q^2 dq}{2\pi^2} P(q) j_\lambda(qs). \quad (15)$$

It is exact for the local line of sight: no plane-parallel or FFT line-of-sight approximation enters. For  $n = n' = 0$  it is the usual  $\sigma_b^2 = \int d^3 s \xi(s) Q_{ww'}(s) / (I_w I_{w'})$ . The pairs of windows needed are  $(W, W)$ ,  $(W, M)$  and  $(M, M)$ , with triples  $(n, n', \lambda) \in \{(0, 0, 0), (0, 2, 2), (2, 0, 2), (2, 2, 0), (2, 2, 2), (2, 2, 4)\}$ . The window  $M$  (kind 'M' in `thecov.Window`, tilde weight  $w_r$ ) is new;  $W$  is the local  $m^2$ . Long modes do not resolve veto holes, so the local window has the right shape, and the holes enter only through  $d_k$ .

**Check** (tested). For a uniform sphere of radius  $R$  seen from far away,  $\hat{\mathbf{x}} \approx \hat{\mathbf{n}}$ ,  $Q_{nn'\lambda}$  vanishes for  $\lambda > 0$ , and Equation 14 gives  $\sigma_{00}^2 = \int q^2 dq / (2\pi^2) P(q) [3j_1(qR)/(qR)]^2$ ,  $\sigma_{22}^2 = \sigma_{00}^2/5$  and  $\sigma_{02}^2 = 0$ . The pair counts reproduce these to 1–3%.

**Resolution in  $\mu$ .** The pair counts evaluate  $S$  at the mean  $\mu = \hat{\mathbf{x}} \cdot \hat{\mathbf{s}}$  of each cell. This is first-order exact, and its second-order error biases  $Q_{224}$  by  $-1\%$  of  $Q_{000}$  with 24 cells. The  $\lambda = 4$  term of Equation 14 carries a larger coefficient than  $\lambda = 0$ , and  $\xi_4$  has no compensating negative tail, so the error became  $-8\%$  in  $\sigma_{22}^2$  for the sphere. With 96 cells it is gone. The SSC therefore keeps its own window library with 96  $\mu$  cells. It also uses fewer near pairs, since small separations matter little for long modes.

## 5 The covariance

Collecting Equation 9 for all multipoles and bins,

$$C_{\ell_1 \ell_2}^{\text{SSC}}(k_i, k_j) = \sum_{X,Y} c_{\ell_1}^X(k_i) c_{\ell_2}^Y(k_j) \sigma_{XY}^2, \quad X, Y \in \{(W, 0), (W, 2), (M, 0), (M, 2)\}, \quad (16)$$

$$c_{\ell}^{(W,n)}(k) = d_k R_{\ell}^{(n)}(k) - \hat{P}_{\ell}(k) g_n, \quad c_{\ell}^{(M,n)}(k) = -\hat{P}_{\ell}(k) g_n, \quad (17)$$

with responses and spectra averaged over each  $k$  bin. The matrix has rank  $\leq 4$ : it is the low-rank term seen in the mocks. Independent caps add with the weights of the combined estimator. For the norm-weighted mean of NGC and SGC,  $C = \sum_r \text{norm}_r^2 C_r / (\sum_r \text{norm}_r)^2$ .

## 6 Numerics

- $\xi_{\lambda}(s)$  on thecov’s  $s$  grid, by direct quadrature on a linear  $q$  grid with  $\Delta q \leq 0.25/s_{\text{max}}$  and a  $1h^{-1}$  Mpc Gaussian damping (irrelevant at the separations that matter).
- Bin averages of  $R_{\ell}^{(n)}$  and  $\hat{P}_{\ell}$  with the Gauss–Legendre nodes of the shell kernels ( $k^2$ -weighted).
- Pair counts: the far pairs of thecov’s subsample ( $n_{\text{sub}}$ ),  $3 \times 10^5$  near pairs, 96  $\mu$  cells. They are saved and reloaded; they depend only on the geometry.
- Cost: a few minutes per cap. The responses and  $\xi_{\lambda}$  take seconds and can be recomputed for any cosmology or bias without new pair counts.

## 7 Validation targets and an estimate

The off-diagonal decomposition of the mock covariance (`rank1_excess.py`) gives quantities that the SSC must predict without fitting:

| LRG1 | $\sigma_{P_0}$ | $\sigma_{P_2}/P_0$ | $r(P_0, P_2)$ | ratio SGC/NGC     |
|------|----------------|--------------------|---------------|-------------------|
| NGC  | 0.51%          | 0.94%              | +0.18         |                   |
| SGC  | 0.66%          | 1.22%              | 0.00          | 1.30 (both modes) |

The ratio 1.30 should follow from the  $\sigma^2$  of the two caps, against  $\sqrt{V_{\text{NGC}}/V_{\text{SGC}}} = 1.39$  for a pure volume scaling. An order-of-magnitude estimate for the isotropic mode of LRG1 NGC: per unit long matter mode, BC gives about +3.5 times  $P_0$  (including dilation at  $n_{\text{eff}} \approx -1.5$ ) and LA gives  $-2g_0 \approx -4.9$ . The net is  $\approx -1.4$ , times  $\sigma_b \approx 0.35\%$  for a volume of  $\approx 1.5 \times 10^9 (h^{-1} \text{Mpc})^3$ . That is  $\sigma_{P_0} \approx 0.5\%$ , the measured order. The near cancellation between BC and LA makes the prediction sensitive to  $b_1$ ,  $b_2$  and the nonlinear response at  $k \gtrsim 0.1$ . This is the main uncertainty of the tree-level model.

## 8 Limitations

- **Tree-level responses.** At  $k \gtrsim 0.1 h \text{Mpc}^{-1}$  the nonlinear response differs from tree level, and fingers of God reduce the redshift-space response. Response coefficients from simulations, or a separate-universe calibration, could replace  $a_{\ell n}$ ,  $c_{\ell n}$  without changing anything else.

- **Bias parameters.**  $b_2$  and  $b_{s^2}$  come from relations ( $b_2(b_1)$ , local Lagrangian), unless they are given. Their effect is largest on  $R_0^{(0)}$  and  $R_2^{(2)}$ .
- **Auto-spectra of one tracer.** Cross-spectra need the responses of two tracers.
- **Not included.** The response of the shot noise; the connected trispectrum of short modes (the part of the excess that grows with the bin width); evolution of the long mode across the redshift range (the linear  $P$  at  $z_{\text{eff}}$  is used).